

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)*

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 25

Code of Conduct

I **Jaivant.S** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **Jaivant.S**

Assessment Tasks

Short Answer Test

SET 1

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

Answers :

1. Define Cloud Computing and Explain Four Characteristics That Make It Different from Traditional On-Premise Computing

Definition of Cloud Computing:

Cloud computing is the delivery of computing services such as servers, storage, databases, networking, software, and analytics over the internet ("the cloud"). Instead of owning and maintaining physical hardware, users can access these resources on demand from a cloud service provider and pay only for what they use.

Four Characteristics of Cloud Computing:

1. On-Demand Self-Service

Users can access computing resources such as storage, virtual machines, and applications whenever they need them without requiring assistance from the service provider. In contrast, traditional on-premise systems often require manual setup and IT support.

2. Scalability and Elasticity

Cloud resources can be quickly increased or decreased based on demand. For example, a business can add more storage or processing power during peak periods and reduce it when demand falls. Traditional on-premise systems require purchasing and installing additional hardware, which takes time and money.

3. Broad Network Access

Cloud services are available over the internet and can be accessed from various devices, including laptops, smartphones, and tablets, from any location with an internet connection. Traditional on-premise systems are often limited to a specific physical location or private network.

4. Pay-as-You-Go (Measured Service)

Cloud computing uses a usage-based pricing model where customers pay only for the resources they consume. This reduces upfront investment and operating costs. In traditional on-premise computing, organizations must

purchase, maintain, and upgrade their own hardware and software regardless of actual usage.

Conclusion:

Cloud computing provides flexible, scalable, and cost-effective computing services over the internet. Its key characteristics—on-demand self-service, scalability, broad network access, and pay-as-you-go pricing—make it significantly different from traditional on-premise computing.

2. Trace the Evolution of Cloud Computing from Hardware Evolution to Internet Software Evolution

Hardware Evolution

- Early computing relied on large **mainframe computers** shared by multiple users.
- Personal computers (PCs) made computing affordable and accessible.
- Powerful servers and networking technologies enabled organizations to build large data centers.

Virtualization Evolution

- Virtualization allowed multiple virtual machines (VMs) to run on a single physical server.
- This improved hardware utilization, reduced costs, and simplified resource management.

Internet Evolution

- High-speed internet made it possible to access computing resources remotely.
- Businesses could deliver applications and services over the web instead of installing them locally.

Software Evolution

- Applications evolved from desktop software to **web-based applications** and **Software as a Service (SaaS)**.
- APIs, automation, and distributed systems enabled scalable cloud applications.

Modern Cloud Platforms

- Cloud providers combine virtualization, high-speed networks, automation, and large-scale data centers.
- This enables users to access computing resources on demand with high availability and scalability.

Conclusion:

The evolution from powerful hardware to virtualization, internet connectivity, and web-based software laid the foundation for modern cloud platforms such as AWS, Azure, and Google Cloud.

3. Explain the Role of Server Virtualization in Cloud Computing. Differentiate Between Virtualization and Cloud Computing

Role of Server Virtualization

Server virtualization divides one physical server into multiple virtual machines (VMs). Each VM runs its own operating system and applications independently.

Benefits:

- Better hardware utilization
- Reduced infrastructure cost
- Easy resource allocation
- Faster deployment
- Improved disaster recovery

Difference Between Virtualization and Cloud Computing

Virtualization	Cloud Computing
Technology that creates virtual machines.	Service model delivering computing resources over the internet.
Focuses on efficient hardware utilization.	Focuses on providing scalable, on-demand services.
Runs on a single physical server.	Uses many virtualized servers across data centers.
Managed within an organization.	Managed by cloud service providers.

Conclusion:

Virtualization is the core technology that enables cloud computing, while cloud computing is the service model built on top of virtualization.

4. Discuss the Role of Data Centers in Delivering Cloud Services

A **data center** is a facility that contains servers, storage devices, networking equipment, cooling systems, and power infrastructure used to deliver cloud services.

Roles of Data Centers

1. **Store Data** – Securely stores customer files, databases, and applications.

2. **Provide Computing Power** – Hosts virtual machines and cloud applications.
3. **Ensure High Availability** – Uses backup servers and redundant power to minimize downtime.
4. **Support Scalability** – Adds more servers as demand increases.
5. **Maintain Security** – Protects data using firewalls, encryption, and physical security.
6. **Enable Disaster Recovery** – Keeps backup copies of data to recover from failures.

Conclusion:

Data centers are the backbone of cloud computing, providing reliable, secure, and scalable infrastructure for cloud services.

5. Differentiate IaaS, PaaS, SaaS, and XaaS with One Real-World Example Each (Point-wise)

1. IaaS (Infrastructure as a Service)

- Provides virtual machines, storage, and networking resources.
- Customer manages the operating system, applications, and data.
- Cloud provider manages the physical hardware and virtualization.
- Suitable for organizations that need maximum control over infrastructure.
- **Example:** Amazon EC2.

2. PaaS (Platform as a Service)

- Provides a platform for developing, testing, and deploying applications.
- Customer manages only the applications and data.
- Cloud provider manages the infrastructure, operating system, and runtime environment.
- Suitable for developers who want to focus on coding without managing hardware.
- **Example:** Google App Engine.

3. SaaS (Software as a Service)

- Provides ready-to-use software over the internet.
- Customer only uses the application.
- Cloud provider manages the application, infrastructure, updates, and maintenance.
- Suitable for end users who need software without installation.
- **Example:** Gmail.

4. XaaS (Anything as a Service)

- Refers to any IT service delivered through the cloud (such as storage, databases, AI, or security).
- Customer management depends on the type of service used.
- Cloud provider manages the service infrastructure.
- Offers flexible cloud-based services for various business needs.
- **Example:** Microsoft 365.

Key Differences

- **IaaS:** Offers maximum control over infrastructure.
- **PaaS:** Focuses on application development without managing hardware.
- **SaaS:** Provides ready-to-use software with no installation or maintenance required.
- **XaaS:** Covers all types of cloud-based services beyond IaaS, PaaS, and SaaS.

Conclusion

- IaaS, PaaS, SaaS, and XaaS provide different levels of control and responsibility.
- Users can choose the service model based on their business requirements, technical expertise, and budget. These models make cloud computing flexible, scalable, and cost-effective.